Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
PI Name:	Williams, James G Ph.D.		
Project Title:	Improving lunar laser ranging tests of gravitational theory		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Jet Propulsion Laboratory		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Turyshev, Slava G. (Jet Propulsion Laboratory) Boggs, Dale H. (Jet Propulsion Laboratory)		
Grant/Contract No.:			
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obsacurec by Earth's gravity		

Task Description:	Precise analyses of accurate ranges to the Moon test several aspects of gravitational theory: the Equivalence Principle, geodetic precession, PPN parameters beta and gamma, and possible variation of the gravitational constant G. The uncertainties of these tests decrease as data accuracies improve and data time span lengthens. This project seeks to improve the Lunar Laser Ranging (LLR) models to exploit the University of California at San Diego/University of Washington APOLLO effort's mm accuracies for improved tests of fundamental and gravitational physics. The goals of this project are three-fold: 1) Develop and implement the multidisciplinary modeling and data analysis improvements necessary to advance LLR from the current cm range accuracy to the expected mm accuracy of the APOLLO effort; 2) Support this new ranging effort, including tests for data accuracy and short- and long-term stability; 3) For future space exploration, investigate the performance characteristics which are desirable for the next generation of laser ranging, the fundamental questions which can be investigated, and the challenges for modeling and data analysis. Precise ranges to the Moon must be accurately analyzed in order to extract the scientific information on the nature of gravity. Accurate models for the computation of the orbit of the Moon and the rotation and orientation of the Earth and Moon, and their realization in computer programs, are needed to do this. This task improves the models and computer programs used for analyzing the Lunar Laser Ranging data. The intent is to improve the accuracy of the data analysis by an order of magnitude in order to match the mm accuracy of the data expected from the new APOLLO facility which is being assembled. The analysis of the highly accurate lunar range data would be used to study the nature of gravitation. Einstein's theory of gravity has been tested by Lunar Laser Ranging and other techniques. This task would improve the accuracy of LLR tests by an order of magnitude. Physics is seeking the next theory of gravity beyond Einstein's theory. The accuracy of the range data is being improved to provide more accurate data on the nature of gravity. An improvement in data accuracy requires an improvement in analysis accuracy if this exploration of gravitation is to be realized.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fundamental research on gravitational physics. Data is also sensitive to geophysical/geodetic parameters. Ephemeris of Moon is publicly available and is used by space missions.
Task Progress:	A description of progress with greater detail can be found in the annual report. In summary: a) One paper titled Improving LLR Tests of Gravitational Theory is published in the International Journal of Modern Physics D. A chapter for the Explanatory Supplement to the American Ephemeris and Nautical Almanac is in press. A third paper on the equivalence principle is in preparation. Two presentations, at Pescara, Italy and Solvang, California, were given. b) Revised input parameters and selection of solution parameters improved fits of LLR data. c) Three improvements were made to the model for computation of lunar orbit and rotation. Two were internal definition changes and the third added Mars torques. d) A new finite difference partial derivatives program was constructed. New partial derivatives files were made. e) Support to the UCSD APOLLO effort was provided by comparing the range predictions from the UCSD prediction program against the JPL analysis program.
Bibliography Type:	Description: (Last Updated: 10/31/2007)
Articles in Peer-reviewed Journals	Williams, J. G.; Turyshev, S. G.; Murphy, Jr., T. W. "Improving LLR Tests of Gravitational Theory" International Journal of Modern Physics D, V13 (No. 3), 567-582 , Mar-2004